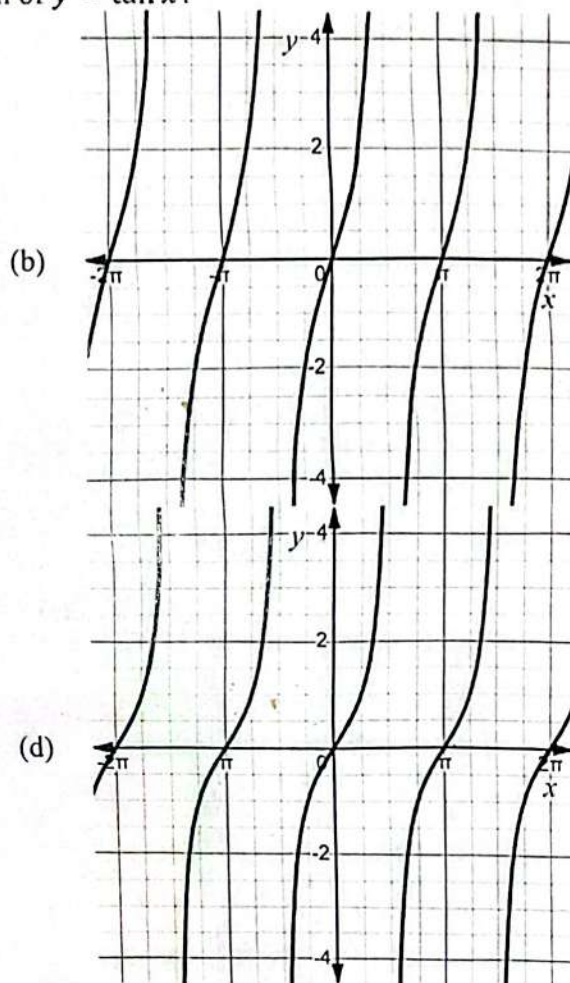
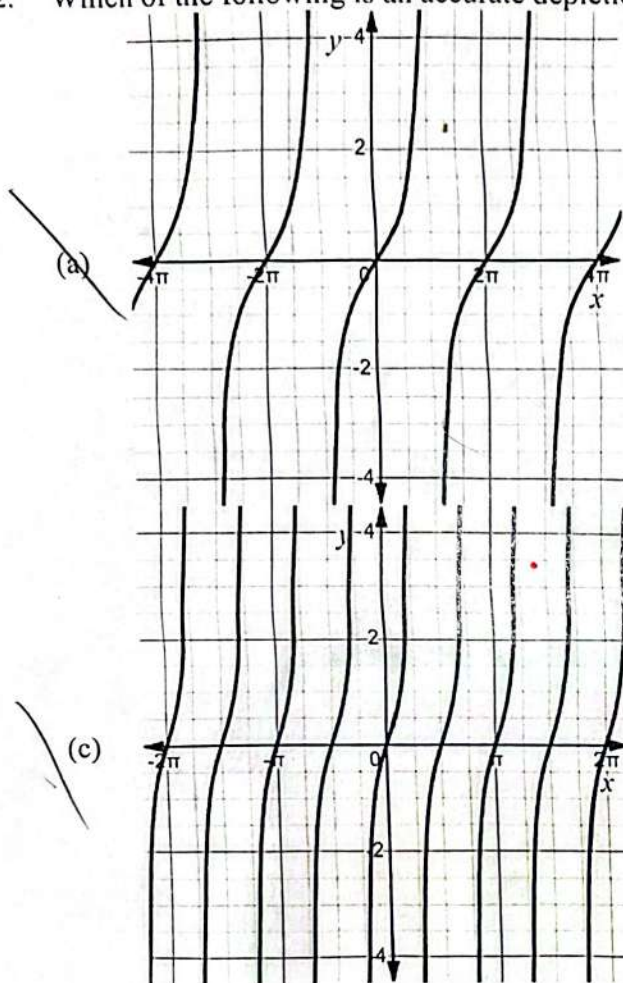


11. Which of the following transformations are true when $y = \cos x$ is transformed into

$$y = 5 \cos \left(-3x + \frac{\pi}{2} \right) + 3$$

- (a) Vertical compression by a factor of 5 (b) Reflection across the x -axis
(c) Horizontal stretch by a factor of 3 (d) Horizontal shift $\frac{\pi}{6}$ units right

12. Which of the following is an accurate depiction of $y = \tan x$?



13. A five-foot fence begins to tilt and the top of the fence is now only four feet from the ground. Determine the obtuse angle that the fence makes with the ground.

- (a) 0.644 (b) 0.927 (c) 2.214 (d) 2.498

14. A point is rotated about a circle. Its start and finish are shown. State the angle.

- (a) $-\frac{3\pi}{2}$ (b) $-\frac{3\pi}{4}$ (c) $-\frac{\pi}{2}$ (d) $-\frac{\pi}{4}$

